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An Experimental Analysis of Brain Stroke Prediction Using Machine Learning Algorithms

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***Abstract*—The brain is a fascinating and complex organ. The skull shields the brain, which consists of the brainstem, cerebellum, and cerebrum. Because stroke is one of the world's most prevalent causes of death, it must be treated quickly to prevent brain damage. A brain stroke can be prevented with early identification, which in turn reduces the mortality rates. One approach is to use machine learning algorithms to identify risk factors. Having a high-quality data collection and cleaning process can streamline the prediction process and help improve the accuracy of predicting brain stroke. This paper proposes a model to achieve an accurate brain stroke forecast. The "Stroke Prediction Dataset" collected from Kaggle was used to train the models. The dataset contains 5110 observations with 12 attributes. Among the seven models used, the gradient-boosting classifier outperformed the rest achieving the highest accuracy of approximately 97.2%. The model can be implemented on cloud-based systems to perform real-time predictions to assist professionals in detecting brain stroke in human beings.**

***Keywords*— Brain Stroke, Gradient Boosting Classifier, Machine Learning**

I. Introduction

Stroke is the third main cause of dysfunction and the second important cause of death, according to the World Health Organization bulletin. Every year, over 15 million individuals have a stroke. Of these, 5 million perish, and further 5 million deal with lifelong, major issues [1, 2]. The symptoms to identify stroke include paralysis of the limbs especially on one side, dizziness, crooked mouth, and difficulty in speaking. In severe cases, the person may even fall into a coma [3]. Even after recovery, many people face problems such as difficulty in speaking, maintaining balance, paralysis, and depression [4, 5]. One of the leading causes of mortality and chronic disability, stroke, emphasizes the value of early risk prediction for prevention and improved patient care. Machine learning has emerged as an important tool in healthcare in recent years, holding promise in predicting brain strokes. The role of machine learning in stroke prediction is examined in this paper, emphasizing its significance, challenges, current research status, and prospects. Timely stroke prediction is crucial for preventing strokes and improving healthcare systems. Several challenges slow the integration of machine learning into stroke prediction. Identifying relevant risk factors and collecting high-quality data pose some of the challenges.

Ensuring that the model is explainable and understandable is crucial for clinical acceptance. Using machine learning to predict brain strokes in healthcare raises ethical concerns. Ensure patient privacy and consent, address bias, and make models transparent. Clarify roles, regularly update, and ensure fair access. Follow regulations, respect patient choices, and consider long-term societal impacts for responsible use. This research paper aims to develop a model that can effectively predict a brain stroke.

II. Literature Review

Several papers have adopted the use of machine learning techniques to classify brain stroke in human beings. Sairam Vasa et al. [6] used SMOTE to resolve the underfitting problem and chose XGB as the best model with an accuracy of 96.34%. Devarakonda et al. [7] obtained a maximum accuracy of 95.70% by using the Random Forest Classifier model. Somya Srivastav et al. [8] split the dataset into 70:30 ratios achieving the highest accuracy of 95.02% using the Logistic Regression Model. KNN was used by Tusher et al. [9] while carefully tuning the value estimate for 'K' and hence the highest accuracy of 97% was achieved. An automatic stroke prediction model based on the CNN method was developed by Chiun-Li, et al. [10]. On a CT scan of the victim's brain, the procedure removed areas that had not been damaged by stroke. The altered images were identified and a technique called Data Augmentation was applied. The image is then trained and evaluated using CNN. Bonna Akter et al. [11] achieved an accuracy of 95.30% using the random forest classifier. Nitya T. Singh et al. [12] compared quantum-enhanced support vector machine algorithms to traditional SVM algorithms and discovered that, with an accuracy of 95.2%, Pegasos QSVC outperforms various classical kernel functions. Manjula et al. [13] compared several algorithms including boosting algorithms such as Light GBM and XGB where XGB outperformed with an accuracy of 97%. Desai et al. [14] achieved 97% accuracy using the logistic regression classifier. Mohapatra et al. [15] used a random forest classifier to achieve 95.16% accuracy. Rashmita Khilar et al. [16] employed support vector machines to provide the highest accuracy of 96.66%.

The existing research showcases several limitations in terms of a lack of thorough data pre-processing and a poor feature selection process. The data cleaning and hyper-parameter process is critical to improving the model's overall accuracy. Lack of thorough cleaning or poor tuning can greatly decrease a model's predictive power. This paper pays high attention to these two processes to reap high accuracies.

III. Motivation And Requirements

In 2016, strokes accounted for 13% of the total deaths in the world [17]. These numbers can be greatly reduced if stroke is caught early. Motivated by the need to build a machine learning model that can save lives by effectively predicting brain stroke, the requirements to achieve effective predictions are highlighted below:

- To carefully address the class imbalance issue and implement a thorough data cleaning.
- To employ quality feature selection and engineering steps to improve data quality.
- To carefully consider the tuning of the models' hyperparameters to improve the predictive accuracy of the algorithms.

The objective is to create a highly predictive and effective model to help predict brain stroke in human beings. Identifying individuals at risk through machine learning not only enhances patient outcomes but also reduces the financial burden on healthcare. Implementing preventive measures, and making stroke prediction is an essential aspect of modern healthcare.

The methodology is covered in section IV, which is broken down into several parts: proposed architecture, data characteristics, data pre-processing, proposed model and comparison with other models, and the metrics used for evaluation. Section V provides the results and discussions followed by the conclusion in section VI.

IV. METHODOLOGY

A. Proposed Architecture

Figure 1 depicts a schematic diagram of the paper's proposed architecture.

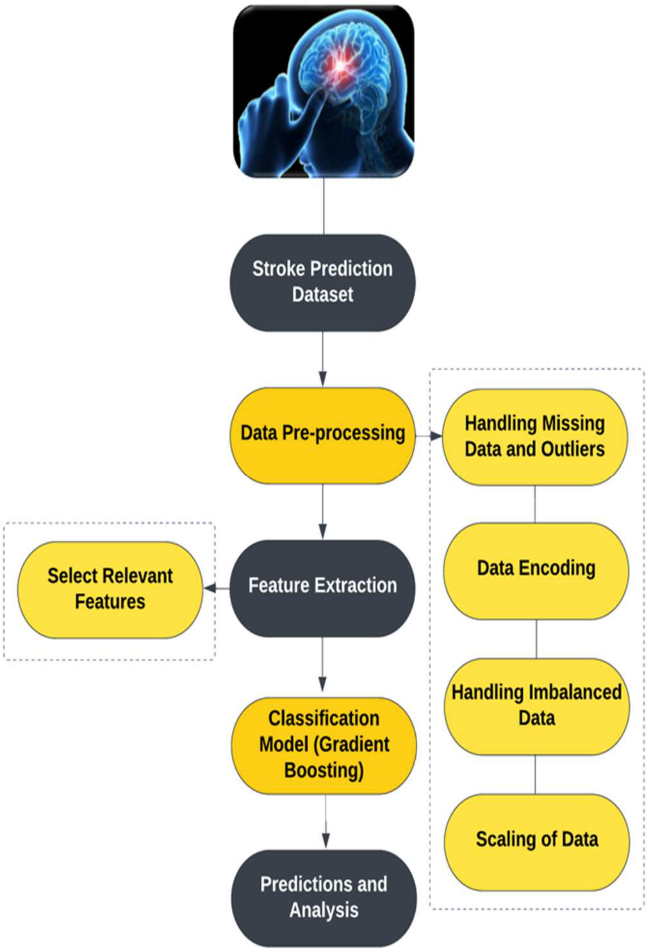

Fig. 1. Schematic Diagram of Proposed Architecture

The training dataset is used to train the models and the testing dataset is used to test the models, and then evaluated based on the evaluation metrics until satisfactory results are obtained. The proposed model can then be used to predict brain stroke in human beings on new unseen data.

B. Dataset Characteristics

Having a good dataset is crucial for efficient research. The dataset utilized is Kaggle's Stroke Prediction Dataset which contains 11 clinical features assisting in the prediction of the occurrences of brain stroke in human beings. The use of the "Stroke Prediction Dataset" from Kaggle provides a foundation for reliable and credible model training, contributing to the robustness of the study. There are 5110 observations alongside 12 attributes in the data. It is an unbalanced dataset, where 4861 patients do not have a stroke while only 249 patients do have a stroke. This dataset is used to forecast the likelihood that a patient would experience a stroke depending on input information. The Patient data is included in each row of the data.

TABLE I. ATTRIBUTES INFORMATION

Sr No.	Attribute Name	Attribute Information
1	id	Distinct identity
2	gender	"Male", "Female" or "Other"
3	age	Patient's age
4	hypertension	1 for having hypertension, else 0
5	heart_disease	1 for having a heart condition, else 0
6	ever_married	"Yes" or "No"
7	work_type [1]	"children", "Private", "Govt_job", "Never_worked", "Self-employed"
8	Residence_type	"Urban" or "Rural"
9	avg_glucose_level	average of the level of glucose in the blood
10	bmi	body mass index
11	smoking_status	"Unknown", "formerly smoked", "never smoked", "smokes"
12	stroke	1 for stroke, else 0

C. Data Pre-processing

The dataset is discovered to have incorrect data types, outliers, class imbalance, and missing values. The missing values in the 'bmi' column are imputed using the column's mean. Irrelevant feature such as 'id' column is dropped. The target feature 'stroke' is separated from the dataset. Label-encoding was employed on specific categorical features such as 'gender', 'ever_married', and 'Residence_type'. One-hot encoding was performed on the entire dataset. The dataset used in the paper is highly imbalanced. Class imbalance has occurred as the minority class labeled 0 as (No) has significantly fewer observations when compared to the majority class labeled 1 as (Yes). For a balanced dataset and to deal with severe class imbalance, SMOTE [19] has been employed which generates synthetic examples to better represent the minority class, allowing the model to make more accurate predictions for the minority class. StandardScaler is used for feature scaling so that the features have 0 mean and a variance of 1. The training set was made up of 80% of the dataset, and 20% of the dataset was used for testing, divided 80:20.

D. Proposed Model

a) Gradient Boosting Classifier

Among the seven models implemented, the Gradient Boosting Classifier is chosen as the proposed model. It creates multiple decision trees and assigns equal weights to all the trees. The first decision tree is trained and the algorithm focuses on the mistakes and then adds another decision tree while focusing on the points that the previous tree got wrong. It uses the gradient descent optimization technique to minimize the errors of the weak models. It also adjusts the weight according to the accuracy of the trees with the help of a loss function and then to obtain the final prediction, computes the weighted average of the predictions from the decision trees. The loss function used is shown in Equ. (1).

$$\text{Loss} = -\frac{1}{\mathrm{N}}\sum_{i=1}^{\mathrm{N}}[y_i \log(p_i) + (1\text{-}y_i) \log(1\text{-}p_i)] \quad (1)$$

E. Comparison With Other Models

a) Logistic Regression

Logistic regression models the relationship between input features and the likelihood of belonging to one of two classes (e.g., 0 or 1). It uses a sigmoid function to convert a linear combination of features into a probability. Model parameters are adjusted during training to reduce prediction errors [20].

b) Gaussian Naïve Bayes

Gaussian Naïve Bayes assumes that features follow a Gaussian (normal) distribution and are independent of each other, hence "naive." It computes the likelihood of an input belonging to a specific class based on the joint probability of its features using Bayes' theorem [21].

c). Decision Tree Classifier

Decision Tree Classifier builds a tree-like structure, and at each split, it tries to create subsets that have mostly one class label. To make predictions, the algorithm employs a set of rules and can handle both numerical and categorical features [22].

d). Extra Trees Classifier

Extra Trees Classifier builds a forest of decision trees. Each tree is fitted to the whole training dataset. When constructing a decision tree, a feature is picked arbitrarily from a subset of available features at each node and the threshold for that selected feature is also chosen randomly and is therefore not based on any optimization criteria like Gini impurity or information gain. The use of random thresholds provides additional randomness to the process. This can lead to a more diverse set of decision trees and can aid in decreasing overfitting and improving the model's generalizability.

e). CatBoost Classifier

CatBoost is a gradient boosting algorithm for classification tasks. It creates decision trees and focuses on the mistakes in the previous trees and thus learns from them. It uses a loss function to identify the trees that are more accurate and assigns higher weights to the ones that are more accurate and lower weights to the trees that are less accurate. The best possible combination of trees is therefore found using the loss function, and the outcome depends on a weighted average of each tree's predictions.

f). XGB Classifier

The XGBoost Classifier uses regularization techniques to prevent overfitting. L1 and L2 regularization techniques are used which works by adding a penalty to the loss function thus preventing it from being too complex. L1 regularization can lead to feature selection. L2 regularization distributes penalty evenly across all features.

$$\mathrm{L_{L1}} = \lambda \sum_{i=1}^{n} |\mathrm{w_i}| \quad (2)$$

$$\mathrm{L_{L2}} = \lambda \sum_{i=1}^{n} \mathrm{w_i^2} \quad (3)$$

F. Hyperparameters Used

TABLE II. HYPERPARAMETER TUNING USED FOR EACH MODEL

Model	Hyperparameters Tuned	Values
LR	-	-
GNB	-	-
DT	criterion	'gini'
ETC	n_estimators	200
	min_samples_split	2
CBC	verbose	False
XGB	n_estimators	500
	learning_rate	0.25
	max_depth	6
	min_samples_split	2
GBC	**n_estimators**	**500**
	learning_rate	**0.25**
	max_depth	**7**

Table 2 provides the hyperparameter tuning used for the implemented models. The hyperparameters are tuned to prevent the models from overfitting or underfitting the data. This is done to prevent the models from capturing noise as well as missing important patterns in the data.

G. Evaluation Metrics Used

Evaluation metrics such as the accuracy, precision, recall, and F1 score are used as evaluation metrics to analyze the models. Accuracy is the ratio of a model's accurate predictions to all of its erroneous predictions. The ratio of successfully estimated positive results among all positive results is referred to as precision. Recall is the ratio of correct positive predictions to all actual positives. The F1 score is the weighted harmonic mean of recall and precision. Figure 2 contains the confusion matrix and describes the relationship between some of the most often-used metrics derived from the confusion matrix.

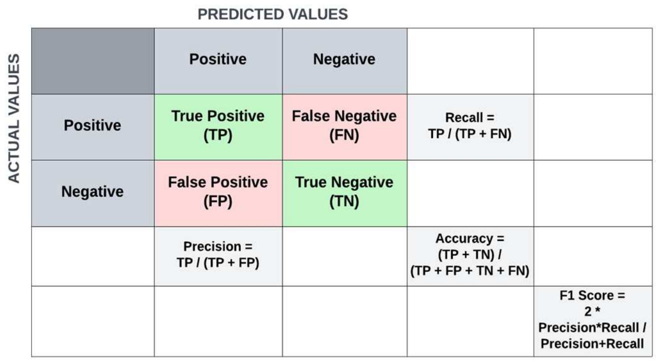

Fig. 2. Confusion matrix

V. RESULTS AND DISCUSSIONS

This section provides a comparison of all the models implemented in this paper with the proposed model. The evaluation metrics used in this paper are used in the comparison. The implemented models are compared on both imbalanced and balanced new, independent test data to provide an insight into the vast improvement in the efficiency of models when used on balanced data compared to when the data is imbalanced.

TABLE III. RESULTS OF IMPLEMENTED MODELS ON NEW IMBALANCED TEST DATA

MODEL	Accuracy (%)	Precision	Recall	F1 Score
LR	72.3	0.12	0.70	0.21
GNB	31.3	0.07	1.00	0.13
DT	85.9	0.07	0.14	0.10
ETC	88.6	0.06	0.09	0.07
CBC	91.1	0.10	0.09	0.09
XGB	91.5	0.16	0.14	0.15
GBC	**93.5**	**0.22**	**0.09**	**0.13**

Table 3 provides the results of all the models implemented on imbalanced data. Gradient Boosting Classifier achieved the highest accuracy (93.5%) on imbalanced data.

TABLE IV. RESULTS OF IMPLEMENTED MODELS ON NEW BALANCED TEST DATA

MODEL	Accuracy (%)	Precision	Recall	F1 Score
LR	92.4	0.96	0.88	0.92
GNB	66	0.59	0.98	0.74
DT	94.2	0.93	0.95	0.94
ETC	96.7	0.97	0.95	0.96
CBC	96.6	0.97	0.96	0.96
XGB	96.3	0.96	0.96	0.96
GBC	**97.2**	**0.97**	**0.96**	**0.97**

Table 4 provides the results of all the models implemented on balanced data. Gradient Boosting Classifier achieved the highest accuracy (97.2%) on balanced data. The accuracy, precision, recall, and F1 score for the Gradient Boosting Classifier are successively 97.2%, 0.97, 0.96, 0.97. The achieved Accuracy was maximum in the Gradient Boosting Classifier (97.2%) followed by the Extra Trees Classifier (96.7%).

TABLE V. CONFUSION MATRIX OF IMPLEMENTED MODELS ON NEW BALANCED TEST DATA

MODEL	Predicted / Actual	No Stroke	Stroke
Logistic Regression	No Stroke	933	33
	Stroke	114	865
Gaussian NB	No Stroke	318	648
	Stroke	14	965
Decision Tree Classifier	No Stroke	902	64
	Stroke	49	930
Extra Trees Classifier	No Stroke	942	24
	Stroke	41	938
Cat Boost Classifier	No Stroke	939	27
	Stroke	39	940
XG Boost Classifier	No Stroke	930	36
	Stroke	36	943
Gradient Boosting Classifier	**No Stroke**	**944**	**22**
	Stroke	**33**	**946**

Table 5 provides the confusion matrix of all the implemented models. It reveals the difference between the outcome predicted by the proposed model and the actual outcome.

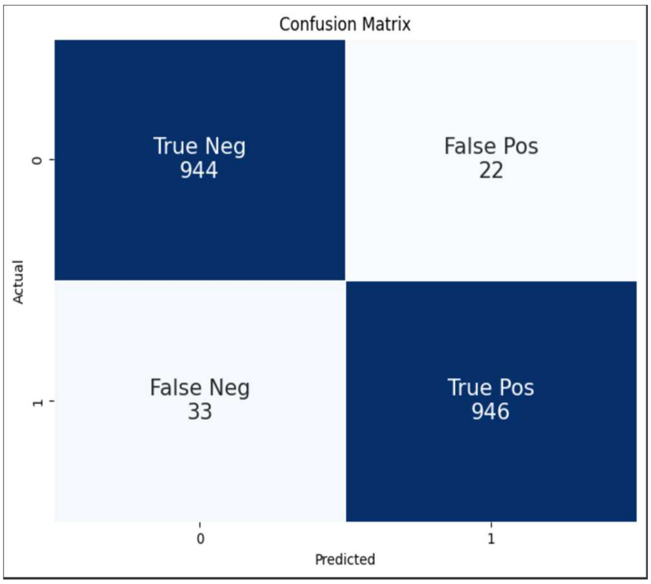

Fig. 3. Confusion matrix of the proposed model on balanced data

The confusion matrix in Figure 3 provides a detailed analysis of the proposed model's performance. Among the seven models, the Gradient Boosting Classifier has 944 TN, 946 TP, 33 FN, and 22 FP. This proves the Gradient Boosting Classifier to be a good model with good predictive power in predicting brain stroke. This research is aimed at finding a model with high potential to classify brain stroke in human beings. Among all the implemented models, the Gradient Boosting Classifier emerged to be the best model for Brain Stroke Prediction with an accuracy of 97.2%.

TABLE VI. COMPARISON OF PROPOSED WORK'S PERFORMANCE WITH EARLIER METHODS

Author	Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Sairam Vasa et al. [6]	XGBoost	96.34	-	-	-
Nagaraju Devarakonda et al. [7]	Random Forest	95.70	-	-	-
Somya Srivastav et al. [8]	Logistic Regression	95.02	92.8	95	92.6
Tusher et al. [9]	KNN	97	97	97	97
Bonna Akter et al. [11]	Random Forest	95.30	-	-	-
Manjula Gururaj Rao et al. [13]	XGBoost	97	96	97	97
Dhyey v. Desai et al. [14]	Logistic Regression	97	95	100	97
Rashmita Khilar et al. [16]	SVM	96.66	97.36	96.1	-
Satapathy et al. [23]	Random Forest	94.50	94	-	-
Proposed	**Gradient Boosting**	**97.2**	**97.7**	**96.6**	**97.2**

VI. CONCLUSIONS

In this study, multiple machine learning classifiers are compared to accurately identify brain stroke. Data cleaning and pre-processing are thoroughly performed and various machine learning techniques such as Standard Scaler, Label Encoding, One-Hot Encoding, and SMOTE are utilized. The future scope can involve developing real-time monitoring systems that continually assess a person's health for stroke risk could revolutionize preventive care. Such systems could be integrated into wearable devices or mobile applications. Our models' dependability and accuracy can be improved by expanding the dataset to include a more balanced dataset by adding more observations, especially for the minority class, and also include a broader range of patient information, such as medical history, genetic markers, and lifestyle factors.

In summary, this study investigated the application of machine learning methods to diagnose brain strokes, and the Gradient Boosting Classifier yielded the best results, with 97.2% accuracy. This result is validated on unseen data and thus provides a good estimate of the model's predictive power. This paper highlights that machine learning can be a valuable tool in the medical field. It has the potential to provide quicker and more accurate diagnoses, which can lead to better care and results for patients.

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